

Module Example Assignment

Chapter X, Sections Y-Z

Your Name

Date Assignment is Due

Problem Set

X.Y.1 Some exercise from the textbook that likely has inline latex like $\int_0^\infty e^{-x} dx$.

The answer to the exercise generally has inline latex as well like $\int_0^\infty e^{-x} dx$. Often times there are lines like the following which are all LaTeX,

$$\begin{aligned}\int_0^\infty e^{-x} dx &= [-e^{-x}]_0^\infty \\ &= \lim_{x \rightarrow \infty} (-e^{-x}) - (-e^0) \\ &= 0 - (-1) = 1.\end{aligned}$$

Software Assignment

Some prompt assigned as a software assignment. For example, create a visualization representing $\int_0^\infty e^{-x} dx$.

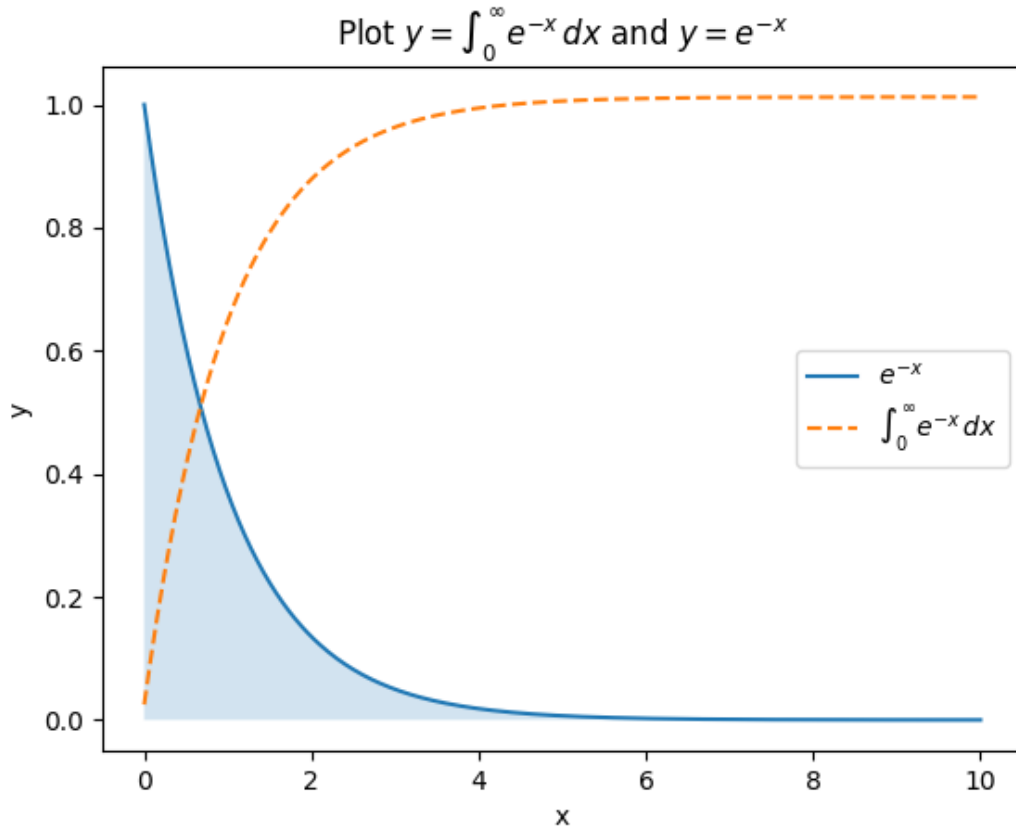


Figure 1: Visualization of e^{-x} and its integral approximation.

Below is the Python code used to generate the figure:

```
import numpy as np
import matplotlib.pyplot as plt

def f(x):
    return np.exp(-x)

x = np.linspace(0, 10, 400)
y = f(x)
integral = np.cumsum(y) * (x[1] - x[0])

plt.plot(x, y, label='$e^{-x}$')
plt.fill_between(x, y, alpha=0.2)
plt.plot(x, integral, label='$\int_0^{\infty} e^{-x} dx$', linestyle='--')
```

```
plt.xlabel('x')
plt.ylabel('y')
plt.title('Plot  $y=\int_0^{\infty} e^{-x} dx$  and  $y=e^{-x}$ ')
plt.legend()
plt.savefig("/workspace/module-example-tex/software-assignment.png")
```